

LEARNING INTENTION

I can describe how weathering, erosion and deposition change Earth's surface over time.

SUCCESS CRITERIA

- ☐ I can define and distinguish weathering and erosion
- ☐ I can give examples of slow and rapid Earth changes
- ☐ I can explain the role of water, wind and ice

KEY CONCEPTS

Weathering: breaking down of rocks (but not moved). **Erosion:** weathered material is picked up and transported. **Deposition:** material is dropped when the transporting agent slows.

WORKED EXAMPLE

Rain hits a cliff (weathering → breaks rock). Water carries the pieces downhill (erosion → transports sediment). Sediment is dropped at the river mouth (deposition → builds a delta).

PART A — WARM UP

1. What is the difference between weathering and erosion?
2. Name THREE agents of erosion.
3. Is the formation of the Grand Canyon a slow or rapid change?
4. What is deposition? Give one example.

PART B — CORE SKILLS

5. Sort into the correct process — weathering, erosion or deposition:
 - Sand building up at a river mouth
 - Wind breaking a rock into small pieces
 - Rain carrying soil down a hillside
6. Give one example of a RAPID change to Earth's surface (not erosion).
7. Explain how plant roots can cause weathering of rocks.
8. Show your working for Question 5 below.

EARTH SCIENCE HELPER

Think: **W-E-T-D** — Weathering breaks, Erosion transports, Transportation moves, Deposition drops. Slow changes happen over thousands of years (rivers carving valleys). Rapid changes happen suddenly (landslides, volcanic eruptions).

WORKING SPACE

PART C — CHALLENGE

9. Real-World Context: After heavy rain near a construction site in Australia, a nearby creek turns muddy brown and sediment builds up at the creek's bend. Using the terms weathering, erosion, transportation and deposition, explain what is happening at each stage.

10. Reasoning: A student says "erosion is always slow." Do you agree? Use ONE example of a slow change and ONE example of a rapid change to support your answer.

REFLECTION

Today I learned...

I CAN CHECKLIST

- ☐ I can define weathering, erosion and deposition
- ☐ I can give slow and rapid change examples
- ☐ I can explain the agents of erosion

TEACHER FEEDBACK

Strength: _____

Next Step: _____

PARENT TIP

After rain, look for soil being washed away in the garden — that's erosion happening right now.

TEACHER TIP

Use a stream table (tray, soil, water) to simulate erosion and deposition — powerful visual experiment.

ANSWER KEY (FOR PARENT / TEACHER USE)

Q1: Weathering = rocks break down in place; erosion = broken material is moved

Q2: Water, wind, ice (accept glaciers)

Q3: Slow (formed over millions of years)

Q4: Deposition = material dropped when transporting agent slows; e.g. sand delta at river mouth

Q5: Sand building up = deposition; wind breaking rock = weathering; rain carrying soil = erosion

Q6: Landslide, volcanic eruption, earthquake, tsunami (accept any one)

Q7: Roots grow into cracks in rocks; as they grow they force the cracks wider until the rock splits

Q8: Working shown as above

Q9: Rain loosens soil (weathering); water picks up and carries sediment downstream (erosion/transportation); sediment settles at bend where water slows (deposition)

Q10: Disagree — slow: river gradually carving a valley over thousands of years; rapid: flood or landslide changing a landscape in minutes